

Mathematics, Algorithms & Data Structures Roadmap

Absolute Beginner → Expert → Research Level

AI / ML / DL

Data Science

Quant Finance

Quantum
Computing

Beginner

Intermediate

Advanced

Expert

Research

18 Chapters 252+ Topics 5 Difficulty Levels 4 Fields

CHAPTER 01

Mathematical Foundations

Core math every practitioner must own before specializing

Topic / Algorithm	Level	Fields	Key Concept / Formula Note
Vectors, Matrices & Tensor Basics	Beginner	AI / ML / DL Data Science Quant Finance Quantum Computing	Operations, norms, broadcasting, Einstein notation
Systems of Linear Equations	Beginner	AI / ML / DL Data Science Quant Finance	Gaussian elimination, row reduction, existence & uniqueness
Eigenvalues & Eigenvectors Eigen	Intermediate	AI / ML / DL Data Science Quant Finance Quantum Computing	Characteristic polynomial, diagonalization, PSD matrices
Singular Value Decomposition SVD	Intermediate	AI / ML / DL Data Science Quant Finance	Low-rank approx, pseudoinverse, condition number, PCA link
Matrix Calculus	Intermediate	AI / ML / DL Data Science Quant Finance	Jacobian, Hessian, gradient of trace/det, chain rule

Topic / Algorithm	Level	Fields	Key Concept / Formula Note
Tensor Algebra & Contraction	Advanced	AI / ML / DL Quantum Computing	Multi-index arrays, reshaping, mode-n products, Tucker
Multivariate Calculus	Intermediate	AI / ML / DL Data Science Quant Finance Quantum Computing	Partial derivatives, gradient, divergence, Laplacian
Taylor Series & Linearization	Intermediate	AI / ML / DL Data Science Quant Finance	Remainder theorem, Newton approx, error bounds
Fourier Transform & FFT FFT	Advanced	AI / ML / DL Data Science Quantum Computing	DFT, spectral analysis, convolution theorem, FFT $O(n \log n)$
Measure Theory & Lebesgue Integration	Advanced	AI / ML / DL Quant Finance Quantum Computing	Sigma-algebras, measurability, L_p spaces, almost-sure conv
Functional Analysis	Advanced	AI / ML / DL Quant Finance Quantum Computing	Banach & Hilbert spaces, bounded operators, dual spaces
Differential Geometry & Manifolds	Advanced	AI / ML / DL Quantum Computing	Tangent spaces, Riemannian metric, geodesics, Lie groups

CHAPTER 02

Probability & Statistics

The language of uncertainty from axioms to advanced theory

Topic / Algorithm	Level	Fields	Key Concept / Formula Note
Probability Axioms & Events	Beginner	AI / ML / DL Data Science Quant Finance Quantum Computing	Kolmogorov axioms, sigma-algebras, set operations
Conditional Probability & Bayes Theorem	Beginner	AI / ML / DL Data Science Quant Finance	Bayes rule, total probability, likelihood ratio
Discrete Distributions	Beginner	AI / ML / DL Data Science Quant Finance Quantum Computing	Bernoulli, Binomial, Poisson, Geometric, Negative Binomial

Topic / Algorithm	Level	Fields	Key Concept / Formula Note
Continuous Distributions	Beginner	AI / ML / DL Data Science Quant Finance Quantum Computing	Gaussian, Exponential, Beta, Gamma, Chi-squared, t, F
Expectation, Variance & Moments	Beginner	AI / ML / DL Data Science Quant Finance	$E[X]$, $\text{Var}(X)$, MGF, CGF, characteristic functions
Law of Large Numbers & CLT	Intermediate	AI / ML / DL Data Science Quant Finance	Weak/strong LLN, CLT convergence rate, Berry-Esseen
Multivariate Gaussian	Intermediate	AI / ML / DL Data Science Quant Finance	Joint density, conditional dist, marginals, Schur complement
Maximum Likelihood Estimation MLE	Intermediate	AI / ML / DL Data Science Quant Finance	Score function, Fisher information, Cramer-Rao bound
Bayesian Inference	Intermediate	AI / ML / DL Data Science Quant Finance	Prior, likelihood, posterior, conjugate priors, MAP
Hypothesis Testing	Intermediate	Data Science Quant Finance	Neyman-Pearson lemma, p-values, power, FDR, Holm-Bonferroni
Information Theory — Entropy & KL	Intermediate	AI / ML / DL Data Science Quantum Computing	$H(X)$, KL, JS, f-divergences, data processing inequality
Exponential Family & Sufficient Stats	Advanced	AI / ML / DL Data Science	Natural params, log-partition function, Pitman-Koopman
Concentration Inequalities	Advanced	AI / ML / DL Data Science	Markov, Chebyshev, Chernoff, Hoeffding, Azuma-Hoeffding
Empirical Process Theory	Research	AI / ML / DL Data Science	Glivenko-Cantelli, uniform LLN, entropy numbers, bracketing
Random Matrix Theory RMT	Research	AI / ML / DL Data Science Quant Finance	Wigner semicircle, Marchenko-Pastur, free probability

CHAPTER 03

Data Structures — Foundations

The building blocks of every algorithm and system at scale

Topic / Algorithm	Level	Fields	Key Concept / Formula Note
Arrays & Dynamic Arrays Array	Beginner	AI / ML / DL Data Science Quant Finance Quantum Computing	Indexing $O(1)$, amortized resize, cache-line layout
Linked Lists LinkedList	Beginner	AI / ML / DL Data Science	Singly/doubly/circular, sentinel nodes, XOR trick
Stacks & Queues Stack	Beginner	AI / ML / DL Data Science Quant Finance	LIFO/FIFO, monotonic stack, deque, sliding window max
Hash Tables & Hash Maps HashMap	Beginner	AI / ML / DL Data Science Quant Finance	Open addressing, chaining, load factor, rolling hash
Binary Trees & BSTs BST	Beginner	AI / ML / DL Data Science	In/pre/post-order, BST invariant, predecessor/successor
Heaps & Priority Queues Heap	Intermediate	AI / ML / DL Data Science Quant Finance	Min/max heap, heapify $O(n)$, k-way merge, lazy deletion
AVL & Red-Black Trees RBTree	Intermediate	AI / ML / DL Data Science	Height balancing, rotations, $O(\log n)$ guaranteed all ops
Tries & Suffix Structures Trie	Intermediate	AI / ML / DL Data Science	Prefix search $O(L)$, Aho-Corasick, suffix array, SA-IS
Segment Trees & Fenwick Trees SegTree	Intermediate	Data Science Quant Finance	Range queries, lazy propagation, point updates $O(\log n)$
Graphs — Adjacency Structures Graph	Intermediate	AI / ML / DL Data Science Quant Finance Quantum Computing	List vs matrix, incidence, compressed sparse row (CSR)
Disjoint Set Union (Union-Find) DSU	Intermediate	AI / ML / DL Data Science	Path compression + union by rank, inverse-Ackermann $O(\alpha(n))$
Skip Lists SkipList	Advanced	Data Science Quant Finance	Probabilistic $O(\log n)$ expected, lock-free variants
Bloom Filters & Count-Min Sketch Bloom	Advanced	AI / ML / DL Data Science	Probabilistic membership, false positive rate, streaming
KD-Trees & Ball Trees KDTree	Advanced	AI / ML / DL Data Science	Nearest neighbor, range search, curse of dimensionality
Persistent Data Structures	Advanced	Data Science Quantum Computing	Path copying, fat nodes, retroactive structures

Topic / Algorithm	Level	Fields	Key Concept / Formula Note
Wavelet Trees WaveletTree	Research	Data Science AI / ML / DL	Rank/select, range frequency, succinct representation

CHAPTER 04 Sorting & Searching Algorithms Fundamental algorithms — backbone of all computing			
Topic / Algorithm	Level	Fields	Key Concept / Formula Note
Comparison Sorts — $O(n^2)$	Beginner	AI / ML / DL Data Science	Bubble, insertion, selection — stability, adaptive input
Merge Sort MergeSort	Beginner	AI / ML / DL Data Science Quant Finance	Divide-and-conquer, stable, external sort, $O(n \log n)$
Quick Sort QuickSort	Beginner	AI / ML / DL Data Science	Lomuto/Hoare partition, randomized pivot, introsort
Heap Sort HeapSort	Intermediate	AI / ML / DL Data Science	In-place $O(n \log n)$, not stable, tournament tree link
Counting, Radix & Bucket Sort RadixSort	Intermediate	AI / ML / DL Data Science	Non-comparison, $O(n+k)$ or $O(nk)$, stable integer sort
Binary Search & Template BinarySearch	Beginner	AI / ML / DL Data Science Quant Finance Quantum Computing	Left/right bisect, search-on-answer pattern, $O(\log n)$
Interpolation & Exponential Search	Intermediate	AI / ML / DL Data Science	$O(\log \log n)$ uniform data, unbounded arrays
Quickselect & Order Statistics Quickselect	Intermediate	AI / ML / DL Data Science Quant Finance	Kth-smallest $O(n)$ avg, median-of-medians $O(n)$ worst
External & Cache-Oblivious Sorting	Advanced	Data Science Quant Finance	k-way merge, I/O complexity, van Emde Boas layout
Parallel & Sorting Networks BitonicSort	Advanced	Quantum Computing AI / ML / DL	Bitonic sort, AKS network, $O(\log^2 n)$ parallel depth

CHAPTER 05 Graph Algorithms Traversal, shortest paths, flows, matching, and spectral theory
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Topic / Algorithm	Level	Fields	Key Concept / Formula Note
BFS & DFS BFS/DFS	Beginner	AI / ML / DL Data Science Quant Finance Quantum Computing	Level-order, connected components, bipartite check
Topological Sort TopoSort	Intermediate	AI / ML / DL Data Science	Kahn BFS, DFS post-order, DAG critical path, scheduling
Dijkstra — SSSP Dijkstra	Intermediate	AI / ML / DL Data Science Quant Finance	Priority queue, $O((V+E) \log V)$, non-negative weights
Bellman-Ford — Negative Edges BellmanFord	Intermediate	AI / ML / DL Data Science Quant Finance	Relaxation $O(VE)$, negative-cycle detection, SPFA variant
Floyd-Warshall — APSP Floyd-Warshall	Intermediate	AI / ML / DL Data Science Quant Finance	$O(V^3)$ DP, transitive closure, dense graphs
A* Search A*	Intermediate	AI / ML / DL Data Science	Heuristic-guided, admissible $h(n)$, IDA* for memory savings
Minimum Spanning Tree MST	Intermediate	AI / ML / DL Data Science Quant Finance	Prim (heap), Kruskal (DSU), Boru?ka, $O(E \log V)$
Max Flow — Push-Relabel & Dinic MaxFlow	Advanced	AI / ML / DL Data Science Quant Finance	Dinic $O(V^2 E)$, push-relabel $O(V^2 \sqrt{E})$, min-cut link
Bipartite Matching — Hopcroft-Karp HopcroftKarp	Advanced	AI / ML / DL Data Science	Max matching $O(E \sqrt{V})$, Hungarian for weighted
SCC — Tarjan & Kosaraju SCC	Advanced	AI / ML / DL Data Science	$O(V+E)$, condensation DAG, 2-SAT reduction
Spectral Graph Theory SpectralGraph	Advanced	AI / ML / DL Data Science Quantum Computing	Graph Laplacian, spectral gap, Cheeger inequality, cuts
Graph Neural Networks — Math Basis GNN	Advanced	AI / ML / DL Data Science	MPNN, WL isomorphism test, spectral vs spatial conv
Random Walks & PageRank PageRank	Advanced	AI / ML / DL Data Science Quant Finance Quantum Computing	Stationary dist, mixing time, personalized PageRank
Quantum Walk Algorithms QuantumWalk	Research	Quantum Computing AI / ML / DL	Szegedy walk, element distinctness $O(N^{2/3})$, MNRS

Dynamic Programming & Recursion

Optimal substructure, memoization, and advanced DP optimization

Topic / Algorithm	Level	Fields	Key Concept / Formula Note
Recursion & Divide-and-Conquer	Beginner	AI / ML / DL Data Science Quantum Computing	Base cases, Master theorem, Akra-Bazzi method
Top-Down DP — Memoization	Beginner	AI / ML / DL Data Science Quant Finance	Recursive + cache table, overlapping subproblems
Bottom-Up DP — Tabulation	Beginner	AI / ML / DL Data Science Quant Finance	Iterative fill, space optimization, rolling array
Knapsack & Subset DP Knapsack	Intermediate	AI / ML / DL Data Science Quant Finance	0/1 & unbounded knapsack, subset sum, partition problem
LCS, Edit Distance & Alignment LCS	Intermediate	AI / ML / DL Data Science	LCS, Levenshtein, Smith-Waterman, affine gap penalty
Matrix Chain & Interval DP IntervalDP	Intermediate	AI / ML / DL Data Science	Optimal parenthesization, palindrome partitioning
Tree DP & Rerooting TreeDP	Intermediate	AI / ML / DL Data Science	Subtree states, rerooting technique, tree diameter
Digit DP DigitDP	Intermediate	AI / ML / DL Data Science	Counting integers with digit constraints, tight-bound trick
Bitmask DP BitmaskDP	Advanced	AI / ML / DL Data Science Quantum Computing	TSP $O(2^n n^2)$, subset enumeration, profile DP
Convex Hull Trick & Li Chao Tree CHT	Advanced	Data Science Quant Finance	$O(n \log n)$ optimization for linear cost functions
Divide-and-Conquer DP Optimization D&C-DP;	Advanced	AI / ML / DL Data Science	$Opt(i,j)$ monotone, $O(n \log n)$ from $O(n^2)$, SMAWK
Knuth-Yao Speedup Knuth	Advanced	AI / ML / DL Data Science	Quadrangle inequality, $O(n^2)$ optimal BST, interval DP
Stochastic DP & Bellman Equations BellmanDP	Advanced	AI / ML / DL Quant Finance	MDP value iteration, policy iteration, Q-function link
Approximate DP ADP	Research	AI / ML / DL Quant Finance	Function approx, ADP, deep RL as ADP, curse of dim

CHAPTER 07

Algorithm Design Paradigms

Greedy, backtracking, randomized, streaming, and approximation

Topic / Algorithm	Level	Fields	Key Concept / Formula Note
Greedy Algorithms Greedy	Beginner	AI / ML / DL Data Science Quant Finance	Activity selection, Huffman, exchange argument proof
Backtracking & Constraint Satisfaction CSP	Intermediate	AI / ML / DL Data Science Quantum Computing	N-queens, Sudoku, arc consistency AC-3, forward checking
Branch and Bound BnB	Advanced	AI / ML / DL Quant Finance	ILP solver, TSP exact, best-first + LP bounding
Randomized Algorithms Randomized	Intermediate	AI / ML / DL Data Science Quant Finance Quantum Computing	Las Vegas vs Monte Carlo, randomized quicksort, Karger
Streaming Algorithms Streaming	Advanced	AI / ML / DL Data Science	Count-Min, HyperLogLog, reservoir sampling, Flajolet-Martin
Online Algorithms & Competitive Ratio Online	Advanced	AI / ML / DL Quant Finance	Ski rental, k-server, paging, primal-dual for online
Approximation Algorithms Approx	Advanced	AI / ML / DL Data Science Quant Finance	FPTAS, PCP theorem, set cover $\ln(n)$, vertex cover 2-approx
Amortized Analysis Amortized	Intermediate	AI / ML / DL Data Science	Potential method, accounting method, dynamic table
Parallel Algorithms — PRAM Model Parallel	Advanced	AI / ML / DL Data Science Quantum Computing	CREW/EREW PRAM, work-depth, parallel prefix $O(\log n)$
External Memory Algorithms ExtMem	Advanced	AI / ML / DL Data Science	I/O model, B-tree, cache-oblivious matrix multiply

CHAPTER 08

Optimization Theory & Algorithms

Continuous and combinatorial optimization for ML, finance, QC

Topic / Algorithm	Level	Fields	Key Concept / Formula Note
Gradient Descent & Convergence GD	Beginner	AI / ML / DL Data Science Quant Finance	GD, SGD, mini-batch, $O(1/k)$ convex, $O(\exp)$ strongly convex
Momentum & Adaptive Optimizers Adam	Intermediate	AI / ML / DL Data Science	Momentum, Nesterov, AdaGrad, RMSProp, Adam, AdamW
Second-Order Methods L-BFGS	Advanced	AI / ML / DL Quant Finance	Newton, L-BFGS, Gauss-Newton, Fisher-Rao natural gradient
Convex Analysis	Intermediate	AI / ML / DL Data Science Quant Finance	Convex sets, supporting hyperplanes, Jensen, epigraph
Lagrangian Duality & KKT Conditions KKT	Advanced	AI / ML / DL Quant Finance Quantum Computing	Primal-dual, strong duality, complementary slackness
Linear Programming LP	Intermediate	AI / ML / DL Data Science Quant Finance	Simplex, revised simplex, LP duality, interior-point
Quadratic Programming QP	Advanced	AI / ML / DL Quant Finance	Active set, interior point, SVM dual, CQP
Integer & Mixed-Integer Programming ILP	Advanced	AI / ML / DL Quant Finance	Branch-and-cut, Gomory cuts, LP relaxation, Lagrangian relax
Semidefinite Programming (SDP) SDP	Advanced	AI / ML / DL Quantum Computing	PSD cone, LMI, SDP for MAX-CUT, quantum state tomography
Stochastic & Variance-Reduced Optimization SVRG	Advanced	AI / ML / DL Quant Finance	SAA, SVRG, SAGA, SARAH, Spider — $O(n + 1/\epsilon^2)$ bounds
Proximal Algorithms & ADMM ADMM	Advanced	AI / ML / DL Data Science Quant Finance	Proximal gradient, FISTA, ADMM, Douglas-Rachford
Frank-Wolfe & Conditional Gradient FW	Advanced	AI / ML / DL Data Science	LMO oracle, sparse updates, $O(1/k)$ rate, away-steps
Mirror Descent & Online Convex Opt MD	Advanced	AI / ML / DL Quant Finance	Bregman divergence, regret $O(\sqrt{T})$, FTRL
Bayesian Optimization BayesOpt	Advanced	AI / ML / DL Data Science Quant Finance	GP surrogate, EI/UCB/Thompson, BO with constraints
Evolutionary Strategies — CMA-ES CMA-ES	Advanced	AI / ML / DL Quant Finance Quantum Computing	Natural ES, CMA-ES, OpenAI-ES, neuroevolution

Topic / Algorithm	Level	Fields	Key Concept / Formula Note
QAOA — Quantum Optimization QAOA	Research	Quantum Computing AI / ML / DL	Cost/mixer Hamiltonian, p-layer ansatz, QAOA landscape

CHAPTER 09

Machine Learning — Core Theory & Algorithms

Statistical learning theory through kernel methods and ensembles

Topic / Algorithm	Level	Fields	Key Concept / Formula Note
Bias-Variance-Noise Decomposition	Beginner	AI / ML / DL Data Science	MSE decomposition, irreducible error, model selection
Linear & Regularized Regression LinReg	Beginner	AI / ML / DL Data Science Quant Finance	OLS, ridge (L2), lasso (L1), elastic net, ADMM for lasso
Logistic Regression & GLMs LogReg	Beginner	AI / ML / DL Data Science	Sigmoid, cross-entropy, Newton-IRLS, Poisson/Gamma GLMs
k-Nearest Neighbors kNN	Beginner	AI / ML / DL Data Science	Distance metrics, cover tree, approximate NN, LSH
Naive Bayes NaiveBayes	Beginner	AI / ML / DL Data Science	Conditional independence, Gaussian/Multinomial/Bernoulli NB
Decision Trees — CART & C4.5 CART	Beginner	AI / ML / DL Data Science	Gini, entropy split, CART, minimal cost-complexity pruning
Bagging & Random Forests RandomForest	Intermediate	AI / ML / DL Data Science Quant Finance	Bootstrap aggregating, feature sampling, OOB error
Gradient Boosting — XGBoost XGBoost	Intermediate	AI / ML / DL Data Science Quant Finance	Functional gradient, shrinkage, XGBoost/LightGBM/CatBoost
Support Vector Machines SVM	Intermediate	AI / ML / DL Data Science	Max-margin, SMO algorithm, kernel trick, multi-class
Kernel Methods & RKHS Kernel	Advanced	AI / ML / DL Data Science	Mercer theorem, kernel PCA, kernel ridge, Nystrom approx
PCA, ICA & Dimensionality Reduction PCA	Intermediate	AI / ML / DL Data Science Quant Finance	SVD-PCA, PPCA, FastICA, sparse PCA, kernel PCA
Manifold Learning UMAP	Advanced	AI / ML / DL Data Science	Isomap, LLE, t-SNE, UMAP, diffusion maps

Topic / Algorithm	Level	Fields	Key Concept / Formula Note
Gaussian Process Regression GPR	Advanced	AI / ML / DL Data Science Quant Finance	GP prior, posterior, marginal likelihood, sparse GP (FITC)
VC Theory & PAC Learning VC	Advanced	AI / ML / DL Data Science	VC dimension, growth function, fundamental theorem of SL
Rademacher & Gaussian Complexity Rad	Research	AI / ML / DL Data Science	Rademacher complexity, localization, fast rates
Online Learning — Perceptron & Winnow Perceptron	Advanced	AI / ML / DL Data Science Quant Finance	Perceptron, Winnow, FTRL, regret bounds, EWA
Meta-Learning & Transfer Theory MetaLearn	Research	AI / ML / DL Data Science	MAML, domain adaptation, task-relatedness bounds

CHAPTER 10

Deep Learning — Architecture Mathematics

Neural networks from backprop to diffusion models and beyond

Topic / Algorithm	Level	Fields	Key Concept / Formula Note
Universal Approximation Theorem UAT	Intermediate	AI / ML / DL Data Science	Width vs depth, ReLU expressivity, Barron's theorem
Backpropagation & Computational Graphs Backprop	Intermediate	AI / ML / DL Data Science	Reverse-mode AD, Jacobian-vector products, autograd
Weight Initialization — Xavier & He	Intermediate	AI / ML / DL Data Science	Variance preservation, dying ReLU, orthogonal init
Batch, Layer & Group Normalization BatchNorm	Intermediate	AI / ML / DL Data Science	Internal covariate shift, train/test discrepancy, LayerNorm
Dropout, L1/L2 & Spectral Norm Dropout	Intermediate	AI / ML / DL Data Science	Bayesian dropout, spectral radius control, weight decay
CNNs — Convolution Mathematics CNN	Intermediate	AI / ML / DL Data Science	Receptive field, equivariance, depthwise, dilated, FFT conv
RNNs, LSTMs & GRUs LSTM	Intermediate	AI / ML / DL Data Science Quant Finance	Vanishing gradient, BPTT, gating, truncated BPTT
Attention — Scaled Dot-Product Attn	Intermediate	AI / ML / DL Data Science	QKV formulation, $O(n^2 d)$, causal masking, RoPE

Topic / Algorithm	Level	Fields	Key Concept / Formula Note
Transformer — Full Mathematics Transformer	Advanced	AI / ML / DL Data Science	Multi-head attention, FFN, pre/post-norm, ALiBi, FlashAttn
Graph Neural Networks GNN	Advanced	AI / ML / DL Data Science Quantum Computing	MPNN, WL-test, PNA, equivariant GNN, EGNN
GAN Theory — Minimax & Divergences GAN	Advanced	AI / ML / DL Data Science	JS divergence, Wasserstein, mode collapse, WGAN-GP
Variational Autoencoders — ELBO VAE	Advanced	AI / ML / DL Data Science	Reparameterization, ELBO, beta-VAE, VQ-VAE, HierVAE
Normalizing Flows NF	Advanced	AI / ML / DL Data Science Quant Finance	Change of variables, Jacobian, RealNVP, Glow, NSF
Score-Based Models & Diffusion Diffusion	Advanced	AI / ML / DL Data Science	DSM, Langevin, DDPM, DDIM, SDE framework, EDM
Neural ODEs & Continuous Dynamics NeuralODE	Advanced	AI / ML / DL Data Science Quant Finance	Adjoint method, CNF, Hamiltonian NN, Lagrangian NN
Energy-Based Models (EBM) EBM	Advanced	AI / ML / DL Data Science	Contrastive divergence, MCMC training, NCSN
Neural Tangent Kernel (NTK) NTK	Research	AI / ML / DL Data Science	Infinite-width GP limit, lazy training, catapult phase
Mechanistic Interpretability Mech	Research	AI / ML / DL Data Science	Circuits, superposition hypothesis, polysemanticity
Scaling Laws — Chinchilla & Beyond ScaleLaw	Research	AI / ML / DL Data Science	Power laws, compute-optimal ratio, emergent abilities

CHAPTER 11

Reinforcement Learning

MDPs, policy optimization, and deep RL from foundations to research

Topic / Algorithm	Level	Fields	Key Concept / Formula Note
Markov Decision Processes MDP	Intermediate	AI / ML / DL Data Science Quant Finance	State, action, reward, transition, discount gamma, POMDP
Bellman Equations Bellman	Intermediate	AI / ML / DL Data Science Quant Finance	V^* , Q^* , Bellman optimality, contraction mapping proof

Topic / Algorithm	Level	Fields	Key Concept / Formula Note
Dynamic Programming for RL DP-RL	Intermediate	AI / ML / DL Data Science Quant Finance	Policy evaluation, policy improvement, GPI theorem
Monte Carlo RL — Prediction & Control MC-RL	Intermediate	AI / ML / DL Data Science Quant Finance	First-visit MC, off-policy MC, importance sampling
TD Learning — TD(0), TD(lambda) TD	Intermediate	AI / ML / DL Data Science Quant Finance	Bootstrapping, eligibility traces, TD(lambda) unification
Q-Learning & SARSA Q-learn	Intermediate	AI / ML / DL Data Science Quant Finance	Off/on-policy, convergence theorem, n-step returns
Deep Q-Networks (DQN) DQN	Intermediate	AI / ML / DL Data Science	Experience replay, target network, double DQN, dueling DQN
Policy Gradient Theorem PG	Advanced	AI / ML / DL Data Science Quant Finance	REINFORCE, baseline, advantage, compatible function approx
Actor-Critic — A3C, A2C, SAC SAC	Advanced	AI / ML / DL Data Science Quant Finance	Entropy regularization, soft value, reparameterized policy
PPO & Trust Region Methods PPO	Advanced	AI / ML / DL Data Science	Clipped surrogate, TRPO, KL constraint, monotone improvement
Distributional RL — C51, QR-DQN C51	Advanced	AI / ML / DL Data Science Quant Finance	Return distribution, Bellman on distributions, Wasserstein
Multi-Agent RL MARL	Advanced	AI / ML / DL Data Science Quant Finance	Nash Q-learning, MADDPG, QMIX, CTDE, emergent comm
Inverse RL & RLHF IRL	Advanced	AI / ML / DL Data Science	MaxEnt IRL, GAIL, reward modeling, Constitutional AI
Model-Based RL — Dyna, MBPO, Dreamer MBRL	Advanced	AI / ML / DL Data Science Quant Finance	World models, Dyna-Q, MBPO, latent space planning, MuZero
RL Theory — Regret & PAC-MDP RLThy	Research	AI / ML / DL Data Science	UCB bandits, Rmax, EULER, gap-dependent bounds
Offline & Conservative RL OfflineRL	Research	AI / ML / DL Data Science Quant Finance	BCQ, CQL, IQL, TD3+BC, distributional shift

CHAPTER 12

Probabilistic Graphical Models

Bayesian networks, MRFs, variational inference, and deep generative

Topic / Algorithm	Level	Fields	Key Concept / Formula Note
Bayesian Networks BayesNet	Intermediate	AI / ML / DL Data Science Quant Finance	DAG factorization, d-separation, exact inference, Naive Bayes
Markov Random Fields MRF	Intermediate	AI / ML / DL Data Science	Clique potentials, Gibbs distribution, Hammersley-Clifford
Variable Elimination VE	Intermediate	AI / ML / DL Data Science	Elimination ordering, treewidth, complexity analysis
Belief Propagation BP	Intermediate	AI / ML / DL Data Science Quantum Computing	Sum-product on trees, loopy BP convergence, BP-decoding
Junction Tree Algorithm JT	Advanced	AI / ML / DL Data Science	Triangulation, clique tree, calibration, max-sum
Hidden Markov Models HMM	Intermediate	AI / ML / DL Data Science Quant Finance	Forward-backward $O(TK^2)$, Viterbi, Baum-Welch EM
Kalman Filter & Variants Kalman	Advanced	AI / ML / DL Data Science Quant Finance	Linear Gaussian, EKF Jacobians, UKF sigma points, particle
Expectation-Maximization EM	Intermediate	AI / ML / DL Data Science Quant Finance	ELBO, E/M-step, GMM, HMM, incomplete-data log-likelihood
Variational Inference & CAVI VI	Advanced	AI / ML / DL Data Science Quant Finance	Mean field, CAVI updates, ELBO gradient, ADVI
Stochastic Variational Inference SVI	Advanced	AI / ML / DL Data Science Quant Finance	Mini-batch ELBO, natural gradient, BBVI, NVIL
Normalizing Flows for Density NF-PGM	Advanced	AI / ML / DL Data Science Quant Finance	Exact likelihood, IAF, MAF, spline flows

CHAPTER 13

Stochastic Processes & Markov Chains

Randomness in time — for ML, RL, sampling, and quant finance

Topic / Algorithm	Level	Fields	Key Concept / Formula Note
Discrete-Time Markov Chains DTMC	Intermediate	AI / ML / DL Data Science Quant Finance Quantum Computing	Transition matrix, Chapman-Kolmogorov, classification
Stationary Distributions & Mixing	Intermediate	AI / ML / DL Data Science Quant Finance	Detailed balance, spectral gap, conductance, Poincare ineq
MCMC — Metropolis-Hastings MCMC	Intermediate	AI / ML / DL Data Science Quant Finance	Detailed balance, geometric ergodicity, acceptance rate
Hamiltonian Monte Carlo HMC	Advanced	AI / ML / DL Data Science Quant Finance	Leapfrog integrator, NUTS sampler, energy conservation
Sequential Monte Carlo SMC	Advanced	AI / ML / DL Data Science Quant Finance	SIS, resampling, particle filter, $O(N)$ per step
Continuous-Time Markov Chains CTMC	Advanced	AI / ML / DL Quant Finance Quantum Computing	Q-matrix, Kolmogorov forward/backward, Poisson process
Brownian Motion & Wiener Process BM	Advanced	Quant Finance AI / ML / DL Quantum Computing	Quadratic variation, nowhere differentiable, Levy's theorem
Stochastic Calculus — Ito Calculus Ito	Advanced	Quant Finance AI / ML / DL	Ito integral, isometry, Ito formula, Girsanov theorem
SDEs — Existence & Numerics SDE	Advanced	Quant Finance AI / ML / DL Quantum Computing	Lipschitz conditions, Euler-Maruyama, Milstein, strong order
Martingales & Optional Stopping Martingale	Advanced	Quant Finance AI / ML / DL	Doob's OST, martingale convergence, de Moivre martingale
Ergodic Theory Ergodic	Advanced	AI / ML / DL Quant Finance Quantum Computing	Birkhoff's ergodic theorem, mixing, time-ensemble equivalence

CHAPTER 14

Monte Carlo & Simulation Methods

Numerical probability, simulation, and computational Bayesian methods

Topic / Algorithm	Level	Fields	Key Concept / Formula Note
Monte Carlo Integration MCInt	Intermediate	AI / ML / DL Data Science Quant Finance Quantum Computing	LLN-based, standard error $O(1/\sqrt{N})$, stratification
Importance Sampling IS	Advanced	AI / ML / DL Data Science Quant Finance	IS weights, variance analysis, self-normalized IS, AIS
Rejection Sampling & Envelope RS	Intermediate	AI / ML / DL Data Science	Proposal envelope, acceptance rate, adaptive rejection
Quasi-Monte Carlo QMC	Advanced	Quant Finance Data Science AI / ML / DL	Halton, Sobol, lattice rules, Koksma-Hlawka, $O(\log^d N/N)$
Variance Reduction VarRed	Advanced	Quant Finance AI / ML / DL Data Science	Control variates, antithetic vars, stratification, MLMC
Monte Carlo Option Pricing LSMC	Advanced	Quant Finance Data Science	GBM simulation, Euler-Maruyama, LSM for American options
MCTS — Monte Carlo Tree Search MCTS	Advanced	AI / ML / DL Data Science	UCT bandit, backprop, MCTS in AlphaGo/MuZero
Simulated Annealing SA	Intermediate	AI / ML / DL Quant Finance Quantum Computing	Metropolis acceptance, cooling schedule, convergence in law
Parallel Tempering PT	Advanced	AI / ML / DL Data Science Quant Finance	Replica exchange, swap acceptance, multimodal sampling
Quantum Monte Carlo QMC-Q	Research	Quantum Computing AI / ML / DL Quant Finance	VMC, DMC, sign problem, fixed-node approximation

CHAPTER 15

Quantitative Finance Mathematics

From asset pricing to risk, derivatives, and algorithmic trading

Topic / Algorithm	Level	Fields	Key Concept / Formula Note
Time Value of Money	Beginner	Quant Finance	Compounding, discounting, annuities, bond math, yield
Modern Portfolio Theory MPT	Intermediate	Quant Finance Data Science AI / ML / DL	Mean-variance, efficient frontier, CML, two-fund separation

Topic / Algorithm	Level	Fields	Key Concept / Formula Note
CAPM & Factor Models CAPM	Intermediate	Quant Finance Data Science AI / ML / DL	Beta, alpha, Fama-French 3/5-factor, APT, factor timing
Black-Scholes Option Pricing BS	Advanced	Quant Finance	Risk-neutral PDE, closed-form, put-call parity, Greeks
Ito Calculus for Finance ItoQF	Advanced	Quant Finance AI / ML / DL	GBM, log-normal returns, stochastic vol SDE, Girsanov
Risk-Neutral Pricing RNP	Advanced	Quant Finance	Equivalent martingale measure, no-arbitrage, replication
Term Structure & Rate Models HJM	Advanced	Quant Finance	Vasicek, CIR, Hull-White, HJM, LIBOR market model
Stochastic Volatility Models Heston	Advanced	Quant Finance Data Science AI / ML / DL	Heston, SABR, rough vol (rBergomi), calibration
GARCH Family GARCH	Advanced	Quant Finance Data Science AI / ML / DL	ARCH, GARCH(1,1), EGARCH, GJR-GARCH, realized vol
VaR, CVaR & Risk Measures VaR	Advanced	Quant Finance Data Science	Historical, parametric, MC VaR, coherent risk measures
Extreme Value Theory EVT	Advanced	Quant Finance Data Science	GEV, GPD, POT method, Hill estimator, tail index
Copulas & Dependence Copula	Advanced	Quant Finance Data Science	Sklar theorem, Gaussian/t/Clayton copula, tail dependence
Market Microstructure HFT	Advanced	Quant Finance AI / ML / DL Data Science	Order book, spread decomposition, Kyle lambda, Glosten-Milgrom
Optimal Execution — Almgren-Chriss OE	Advanced	Quant Finance AI / ML / DL	Temporary/permanent impact, TWAP, DP for liquidation
Statistical Arbitrage StatArb	Advanced	Quant Finance Data Science AI / ML / DL	Cointegration, pairs trading, Kalman spread, ML alpha
RL for Quant Finance RL-QF	Advanced	Quant Finance AI / ML / DL Data Science	Portfolio optimization RL, mean-variance RL, deep hedging
Persistent Homology in Finance TDA	Research	Quant Finance AI / ML / DL Data Science	TDA, Betti numbers, persistence diagrams, market regimes

Quantum Computing — Math & Algorithms

From qubit algebra and quantum circuits to QML and fault tolerance

Topic / Algorithm	Level	Fields	Key Concept / Formula Note
Complex Vector Spaces & Dirac Notation Dirac	Intermediate	Quantum Computing AI / ML / DL	Ket/bra, inner product, Hilbert space, dual space
Qubits & Bloch Sphere Qubit	Intermediate	Quantum Computing	Superposition, computational basis, polar/azimuthal angles
Quantum Gates — Single & Multi-Qubit QGates	Intermediate	Quantum Computing AI / ML / DL	Pauli, H, T, CNOT, Toffoli, universality, gate synthesis
Quantum Circuit Model QCircuit	Intermediate	Quantum Computing AI / ML / DL	Circuit depth, T-count, Clifford+T, gate decomposition
Tensor Products & Entanglement TensorQ	Intermediate	Quantum Computing AI / ML / DL	Kronecker product, Bell states, Schmidt decomposition
Density Matrices & Mixed States DensityMat	Advanced	Quantum Computing AI / ML / DL	ρ , purity $\text{tr}(\rho^2)$, partial trace, quantum channels
Quantum Measurement & Born Rule Measure	Intermediate	Quantum Computing AI / ML / DL	Projective, POVM, indirect measurement, quantum Zeno
Quantum Fourier Transform QFT	Advanced	Quantum Computing AI / ML / DL	$O(n^2)$ circuit, inverse QFT, phase kickback, period finding
Quantum Phase Estimation QPE	Advanced	Quantum Computing AI / ML / DL	Ancilla register, Hadamard test, eigenvalue readout
Shor Factoring Algorithm Shor	Advanced	Quantum Computing AI / ML / DL	Order finding via QPE+QFT, CRT reconstruction, RSA threat
Grover Search Algorithm Grover	Advanced	Quantum Computing AI / ML / DL	Oracle+diffusion, $O(\sqrt{N})$, optimal query lower bound
Variational Quantum Eigensolver VQE	Advanced	Quantum Computing AI / ML / DL	Ansatz, UCCSD, parameter shift rule, VQE landscape
QAOA QAOA	Advanced	Quantum Computing AI / ML / DL Quant Finance	Cost/mixer unitaries, p-layer, MaxCut, portfolio optimization

Topic / Algorithm	Level	Fields	Key Concept / Formula Note
Quantum Machine Learning QML	Advanced	Quantum Computing AI / ML / DL	Quantum kernels, QNN, dequantization, quantum advantage
HHL — Linear Systems Algorithm HHL	Advanced	Quantum Computing AI / ML / DL Quant Finance	Quantum matrix inversion, $O(\log N)$, condition number cost
Quantum Error Correction QEC	Expert	Quantum Computing	Shor code, CSS codes, stabilizer formalism, threshold thm
Topological Quantum Computation TQC	Research	Quantum Computing	Anyons, braiding, Majorana zero modes, fault tolerance
Quantum Complexity — BQP & QMA BQP	Research	Quantum Computing AI / ML / DL	BQP vs P, QMA-complete problems, quantum PCP conjecture
Quantum Random Walks QRW	Research	Quantum Computing AI / ML / DL	Coined/Szegedy, $O(N^{2/3})$ element distinctness, NAND trees
Quantum Simulation QSim	Research	Quantum Computing AI / ML / DL Quant Finance	Trotter-Suzuki, qubitization, LCU, electronic structure

CHAPTER 17

Information Theory & Coding

Shannon theory, data compression, channels, and quantum information

Topic / Algorithm	Level	Fields	Key Concept / Formula Note
Shannon Entropy	Intermediate	AI / ML / DL Data Science Quantum Computing	$H(X)$, self-information, entropy of mixtures, max-entropy
Joint, Conditional & Relative Entropy	Intermediate	AI / ML / DL Data Science	$H(X,Y)$, $H(X Y)$, chain rule, KL divergence properties
Mutual Information & Data Processing MI	Intermediate	AI / ML / DL Data Science Quant Finance Quantum Computing	$I(X;Y)$, DPI, Markov chains, sufficient statistics
f-Divergences & Integral Probability fdiv	Intermediate	AI / ML / DL Data Science Quant Finance	KL, JS, Total Variation, Wasserstein, chi-squared

Topic / Algorithm	Level	Fields	Key Concept / Formula Note
Channel Capacity — Shannon Theorem ChannelCap	Advanced	AI / ML / DL Data Science Quantum Computing	BSC, BEC, Gaussian channel, coding theorem, converse
Rate-Distortion Theory RDT	Advanced	AI / ML / DL Data Science	D(R) function, Blahut-Arimoto algorithm, semantic coding
MDL & Bayesian Model Selection MDL	Advanced	AI / ML / DL Data Science	Two-part code, NML, BIC, stochastic complexity
Huffman & Arithmetic Coding	Intermediate	AI / ML / DL Data Science	Optimal prefix codes, arithmetic coding, asymmetric NS
Lempel-Ziv & Universal Compression LZ	Advanced	AI / ML / DL Data Science	LZ77/78, LZW, asymptotic optimality, entropy rate
Error-Correcting Codes ECC	Advanced	AI / ML / DL Data Science Quantum Computing	Hamming, Reed-Solomon, Turbo, LDPC, polar codes, capacity
von Neumann Entropy & Quantum Info VNE	Advanced	Quantum Computing AI / ML / DL	S(rho), quantum channels, coherent info, entanglement dist
Holevo Bound Holevo	Advanced	Quantum Computing	Classical capacity of quantum channel, accessible info

CHAPTER 18

Complexity Theory & Computability

Complexity classes, lower bounds, and foundations of algorithms research

Topic / Algorithm	Level	Fields	Key Concept / Formula Note
Asymptotic Notation & Recurrences BigO	Beginner	AI / ML / DL Data Science Quant Finance Quantum Computing	Big-O/Omega/Theta, Master theorem, Akra-Bazzi, generating fn
Complexity Classes — P, NP, PSPACE Complexity	Intermediate	AI / ML / DL Data Science Quantum Computing	Deterministic/nondeterministic TMs, certificate, oracle
NP-Completeness & Reductions NPC	Intermediate	AI / ML / DL Data Science Quant Finance Quantum Computing	Cook-Levin, Karp reductions, 3-SAT, independent set

Topic / Algorithm	Level	Fields	Key Concept / Formula Note
Approximation Hardness — PCP Theorem PCP	Advanced	AI / ML / DL Data Science Quant Finance	PCP theorem, inapproximability of MAX-3SAT, vertex cover
Fixed-Parameter Tractability FPT	Advanced	AI / ML / DL Data Science	FPT, W-hierarchy, treewidth, kernelization, color coding
Circuit Complexity Circuit	Advanced	AI / ML / DL Data Science Quantum Computing	Boolean circuits, AC0 lower bounds, TC0, NC hierarchy
Communication Complexity CommComp	Advanced	AI / ML / DL Data Science Quantum Computing	Deterministic/randomized CC, corruption bound, XOR lemma
Fine-Grained Complexity FineGrain	Advanced	AI / ML / DL Data Science	SETH, 3SUM, OV, conditional lower bounds, matching theory
Average-Case Complexity AvgCase	Advanced	AI / ML / DL Data Science Quantum Computing	Distributional problems, one-way functions, planted clique
Quantum Complexity — BQP & QMA BQPQMA	Advanced	Quantum Computing AI / ML / DL	Quantum query lower bounds, QMA-completeness, oracle sep
Kolmogorov Complexity & Randomness KC	Advanced	AI / ML / DL Data Science	K-complexity, incompressibility, Martin-Lof randomness
Cryptographic Hardness CryptoHard	Advanced	AI / ML / DL Quantum Computing Quant Finance	OWF, DDH, LWE, SIS, post-quantum assumptions